

SECTION A

1 (long)

(a) State Thevenin's theorem. Explain how *loop current analysis* can be used to determine currents and voltages in an electrical circuit. [7]

(b) Figure 1(a) shows the circuit for an a.c. bridge measurement system. The voltage source is an a.c. source of frequency ω . When the bridge is balanced, the current through the meter M is zero. Find the conditions for balance in terms of the resistances R_1 , R_2 , R_3 , R_4 and the ratio L/C . [15]

(c) Figure 1(b) shows a second bridge circuit, a Wien bridge. Find the frequency at which the bridge balances in terms of R_3 , R_4 and C . [8]

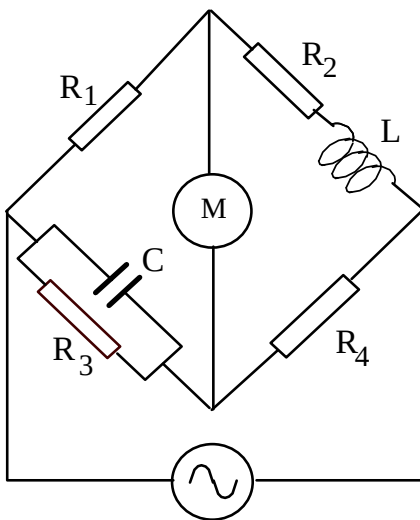


Fig. 1(a)

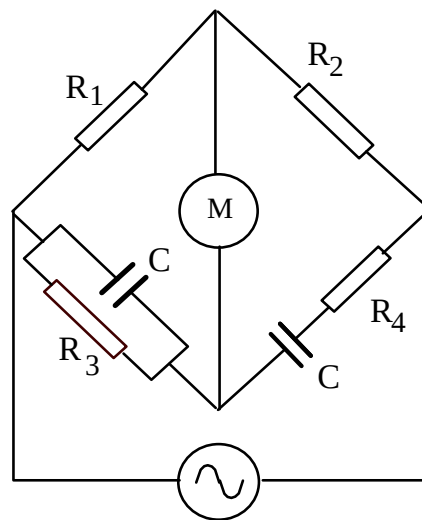


Fig. 1(b)